

ARNA GHOSH

Computer Science, McGill University & MILA
Montréal, Canada
Email ◇ GitHub ◇ LinkedIn ◇ Scholar ◇ Twitter

EDUCATION

McGill University, Montréal
Ph.D. Computer Science

September 2019 - Present
CGPA: 4.0/4.0

McGill University, Montréal
M.Sc. Neuroscience

September 2017 - August 2019
CGPA: 4.0/4.0

Indian Institute of Technology, Kharagpur
B.Tech. Electrical Engineering
Minor in Computer Science
Micro-specialization in Embedded Wireless Technology

July 2013 - May 2017
CGPA: 9.26/10.0

INTERNSHIPS AND WORK EXPERIENCE

Scientist in Residence, Innodem Neurosciences

June 2020 - Present

- Developed a well-documented **Tensorflow** module to implement the *Deepdream* algorithm in order to visualize features extracted by eye gaze-prediction neural networks
- Advised the software team on architecture modifications and further steps based on visualization results

Developer, CBrain, McGill Centre for Integrative Neuroscience

July 2019 - August 2019

- Designed a well-documented **Python** wrapper library around the CBrain web platform to run neuroimaging tasks on a high-performance cluster from a UNIX terminal

MITACS Globalink Research intern, McGill University

May 2016 - July 2016

Mechanism underlying sensorimotor integration during motor learning
Dr. Marie-Hélène Boudrias

- Designed the software and hardware interface for tasks to study motor learning characteristics under different external stimulation conditions using **Python**, **C++** and **OpenCV**
- Interfaced EEG, EMG and fMRI setups to the designed software for different experiments. [Media]

Summer intern, University of British Columbia Okanagan

May 2015 - July 2015

Slide-Scanning Microscopy on a Smart Phone by High-Frame-Rate Video Capture
Dr. Kenneth Chau

- Developed the image processing module for fast deblurring of microscopic images extracted from video and developed an integrated hardware-software interface using an Arduino

MAJOR PROJECTS

DeepMouse

Sep 2019 - Present

Doctoral thesis project
Dr. Blake Richards

- Working on biologically-plausible deep learning models of the visual cortex, with special focus on **contrastive learning** models.
- Building deep learning framework to predict the optimal stimuli and estimate receptive field of mice neurons at different stages of visual information processing.

Deep Learning for Neuroimaging data analysis

Sep 2017 - Aug 2019

Masters thesis project

Dr. Marie-Hélène Boudrias & Dr. Georgios D. Mitsis

- Developed a DL framework in **Torch** to identify exercise-induced task-EEG signatures.
- Developed **ccCAM** (cue-combination for Class Activation Map) to improve network interpretability.
- Designed a ML-based **brain age prediction** system from MRI and MEG recordings in **PyTorch**.
- Developed a **Graph-CNN** approach for analyzing functional connectivity data to predict brain age and compared to other ML-based approaches.

Brexting: Brain Texting

Dec 2017 - Aug 2018

Intel Innovate FPGA Grand Finalists

- Built a DL framework on an Intel FPGA board for decoding of imagined movements from EEG recordings to enable real-time typing.
- Won the **Silver award** in regional finals and **Iron award** in grand finals. Code available on repo

Deep Learning for mitotic figure detection

Jul 2016 - May 2017

Bachelor's Thesis Project Dr. Debdoot Sheet

- Built a DL framework to localize and identify mitotic nuclei from *breast histopathological* images on **Torch** and extended the idea to *leukemia detection*.
- Awarded the **Systems Society award** for best project work on Systems.

IndicView

Oct 2014 - Dec 2015

Google-IIT Pilot Project

Dr. Pawan Goyal

- Implemented a binarization algorithm to segregate text from background and extract words from a document written in an Indian language. Accepted to be integrated into **Google Translate**.

Autonomous Ground Vehicle Research Group

Mar 2014 - May 2017

Perception Team

Dr. Debashish Chakravarty

- Developed image processing modules to detect obstacles, detect lanes and construct a world map by fusing data from camera and LIDAR on **Robot Operating System (ROS)**

PUBLICATIONS

Full Papers and Preprints:

1. A. Xifra-Porxas*, **A. Ghosh***, G.D. Mitsis, M.H. Boudrias, *Estimating brain age from structural MRI and MEG data: Insights from dimensionality reduction techniques*, BiorXiv, Nov 2019. [Link](#)
2. **A. Ghosh**, F. Dal Maso, M. Roig, G.D. Mitsis, M.H. Boudrias, *Unfolding the effects of acute cardiovascular exercise on neural correlates of motor learning using Convolutional Neural Networks*, *Frontiers in Neuroscience* 13 (2019): 1215. [Link](#)
3. E. Renda, S.A. Karmali, I. Yordanova, S. Schwartz, Y. Mahdid, **A. Ghosh**, S. Blain-Moraes, M.H. Boudrias, *Effect of Transcranial Direct Current Stimulation on an Individuals Ability to Learn to Control a Brain-Computer Interface*, *McGill Journal of Medicine*, October 2019. [Link](#)
4. **A Ghosh**, S Singh, D Sheet, *Simultaneous Localization and Classification of Acute Lymphoblastic Leukemic Cells in Peripheral Blood Smears Using a Deep Convolutional Network with Average Pooling Layer*, *IEEE 12th International Conference on Industrial and Information Systems (ICIIS)*, Dec 2017. [Link](#)

*Equal contribution

5. V Mohanty, S Agrawal, S Datta, **A Ghosh**, V D Sharma, D Chakravarty, *DeepVO: A Deep Learning approach for Monocular Visual Odometry*, ArXiv, Nov 2016 [Link](#)

Poster Presentations:

1. A.Kemtur*, **A. Ghosh***, Arthur Dehgan, Tarek Lajnef, Raphael Vallat, Jean-Baptiste Eichenlaub, Perrine Ruby, Karim Jerbi, *Unraveling the neural signatures of dream recall in EEG: a deep learning approach*, Montreal Artificial Intelligence & Neuroscience, November 2019.
2. **A. Ghosh***, A. Xifra-Porxas*, G.D. Mitsis, M.H. Boudrias. *Combining structural MRI images and MEG recordings for Biological brain age prediction*, Organization of Human Brain Mapping (OHBM), Annual Meeting, June 2019. [Link](#)
3. **A. Ghosh***, A. Xifra-Porxas*, G.D. Mitsis, M.H. Boudrias. *Biological brain age prediction using structural MRI: Insights from dimensionality reduction techniques*, International Society for Magnetic Resonance in Medicine (ISMRM), Annual Meeting, May 2019. [Link](#)
4. **A. Ghosh**, F. Dal Maso, G.D. Mitsis, M.H. Boudrias, *Deep Convolutional Architecture with discriminative feature visualization for analysis of EEG data*, Neuroinformatics, August 2018.
5. **A. Ghosh**, Arthur Dehgan, Tarek Lajnef, Raphael Vallat, Jean-Baptiste Eichenlaub, Perrine Ruby, Karim Jerbi, *Unraveling the neural correlates of dream recall: Novel insights from deep convolutional nets*, Neuroinformatics, August 2018.
6. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *AdGAP: Advanced Global Average Pooling*, AAAI-18 Student Abstract and Poster Program (SA-18), February 2018.
7. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *Training Autoencoders in Sparse Domain*, AAAI-18 Student Abstract and Poster Program (SA-18), February 2018.
8. **A. Ghosh**, F. Dal Maso, G.D. Mitsis, M.H. Boudrias, *Deep Semantic Architecture for analysis of motor task EEG data*, Montreal Artificial Intelligence & Neuroscience, November 2017. [**Facebook Best poster award**]
9. F. Dal Maso, B. Desormeau, **A. Ghosh**, M. Roig, M.H. Boudrias, *An acute bout of exercise performed immediately after motor learning alters Beta-band oscillations in the motor cortex*, The Society for Neuroscience, 47th Annual Meeting, November 2017.
10. M. Kassanopoulos, **A. Ghosh**, E. Dhamala, M.H. Boudrias, G.D. Mitsis. *Cardiac Noise Removal from BOLD fMRI based on a Dynamic Linear Model*, Organization of Human Brain Mapping (OHBM), Annual Meeting, June 2017.
11. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *Handwriting Profiling using Generative Adversarial Networks*, AAAI-17 Student Abstract and Poster Program (SA-17), February 2017. [Link](#)
12. **A Ghosh**, B Bhattacharya, S B R Chowdhury, *SAD-GAN: Synthetic Autonomous Driving using Generative Adversarial Networks*, 30th Annual Conference on Neural Information Processing Systems (NeurIPS) 2016: Deep Learning for Action and Interaction, December 2016.[Link](#)
13. K K Agrawal, **A Ghosh**, *Generative Adversarial Motion to Image Synthesis*, 14th European Conference on Computer Vision (ECCV) 2016 Workshop on Biological and Artificial Vision, Amsterdam, October 2016.

TEACHING AND VOLUNTEERING EXPERIENCE

TA, Applied Machine Learning (COMP-551)

Sep - Dec 2020

Dr. Siamak Ravanbakhsh, McGill University

Lab Representative, Member of student body

Aug 2020 - Present

Montreal Institute for Learning Algorithms (MILA)

*Equal contribution

Co-organizer, Neural Information Processing Systems Montreal meetup <i>Montreal Institute for Learning Algorithms (MILA)</i>	Dec 2019
TA, Computational Perception (COMP-546) <i>Dr. Michael Langer, McGill University</i>	Jan - Apr 2019, 2020
Reviewer, Reproducibility Challenge <i>International Conference on Learning Representations (ICLR)</i>	Jan 2019
Co-organizer, Deep Learning and Neuroscience reading group <i>Montreal Neurological Institute (MNI)</i>	Dec 2018 - Present
Student Volunteer, HBHL Symposium and Research Day <i>Healthy Brains for Healthy Lives (HBHL), McGill University</i>	9-10 May 2018
Tutor, Deep Learning for Visual Computing Summer School <i>Dr. Debdeep Sheet, IIT Kharagpur</i>	2-8 Jul 2017

SCHOLARSHIPS AND AWARDS

- Healthy Brains for Healthy Lives (**HBHL**) Doctoral fellowship, September 2020 (\$15,000)
- Quebec Bio-Imaging Network (**QBIN**) travel award, June 2019 (\$500)
- 2nd prize in McGill **TechIdea** Pitch Competition during the annual John D. Thompson Entrepreneurial Development Seminar: presented “*Telesphorus - brain age prediction system*”, January 2019 (\$250)
- Healthy Brains for Healthy Lives (**HBHL**) Masters fellowship, September 2018 (\$10,000)
- McGill **Faculty of Medicine** Internal fellowship, September 2018 (\$10,000)
- Quebec Bio-Imaging Network (**QBIN**) Foreign Students Scholarship, December 2017 (\$7,000)
- Integrated Program in Neuroscience (**IPN**) Recruitment award 2017 (\$5,000)
- **MITACS** Graduate Fellowship award 2017 (\$15,000) for pursuing graduate studies in Canada
- **MITACS** Globalink Research Internship award 2016 (\$7,500) for summer internship 2016
- **Honda** Young Engineer and Scientist award, January 2016 (\$3,000)
- **O.P. Jindal** Engineering & Management Scholarship for 2015 & 2016 (\$3,200)
- **Facebook-MAIN** Best Poster award, Montreal AI & Neuroscience Conference, Nov 2017
- **Systems Society Award** for best project work in 8th semester on Systems, May 2017
- Institute **Silver medal** for securing highest CGPA among Electrical engineering graduates, 2017

EXTRA-CIRRICULAR ACTIVITIES

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| 🏆 Best All-rounder Outgoing student (2017) | 🎤 Elocution - 2 Gold, 3 Silver (2014-17) |
| 🎭 Governor , Bengali Dramatics Society (2016-17) | ⚽ Field Hockey - 1 Bronze (2016) |
| 🗣️ Debate - 1 Gold, 1 Silver (2013-15) | 💻 IOT-based Hackathon - Gold (2015) |
| 🎭 Dramatics - 10 plays, 2 Gold, 4 Silver (2013-17) | 👤 Director , IIT Kharagpur MUN (2014) |
| 🤖 Robot Maze Traversal event - Best Freshers (2014) | 🏆 Best All-rounder fresher (2014-15) |